Homework on SQL Query

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Homework:

Section A – Database and Table Creation

1. Create a Database

Write a MySQL command to create a database named:
companydb
Then write a command to select companydb for use.

2. Create an Employee Table

Inside the companydb database, create a table named employee with the following columns:

Column	**Data Type**	**Constraint**
EMPNO	INT	Primary Key
ENAME	VARCHAR(30)	Not Null
SALARY	DECIMAL(10,2)	–
AGE	INT	–
DEPTID	VARCHAR(10)	–

3. Create a Department Table

Create a table named department with the following structure:

Column	**Data Type**	**Constraint**
DEPTID	VARCHAR(10)	Primary Key
DEPTNAME	VARCHAR(30)	Not Null
LOCATION	VARCHAR(30)	–

4. Create a Student Table

Create a table named student with the following columns:

Column	Data Type
ROLLNO	INT
NAME	VARCHAR(40)
AGE	INT
MARKS	DECIMAL(5,2)
CITY	VARCHAR(30)

Make ROLLNO the primary key.

Section B – Relational Operators

Assume the following employee table contains employee records.

5. Equality Operator

Write a query to display all employees who belong to department:

D001

Use the equality operator =.

6. Greater Than Operator

Write a query to display the employee number, employee name, and salary of employees whose salary is **greater than 15000**.

Use:

>

7. Less Than or Equal To Operator

Write a query to display all employees whose age is **less than or equal to 30**.

Use:

<=

8. Not Equal Operator
Write a query to display all employees who **do not belong to department D002.**
Use either:
<>
or:
!=

Section C – BETWEEN Clause
9. Salary Range
Write a query to display all employees whose salary is between:
10000 and 16000
Use the BETWEEN operator.

10. Age Range
Write a query to display the employee number, employee name, and age of employees whose age is between **25 and 35**, inclusive.

11. Employee Number Range
Write a query to display all employees whose employee number is between:
1003 and 1009
Use the BETWEEN clause.

Section D – Logical Operators
12. Using AND
Write a query to display employees who satisfy **both** conditions:
- Salary is greater than 12000
- Department is D001

Use the AND operator.

13. Using OR
Write a query to display employees who belong to either:
D001
or
D002
Use the OR operator.

14. Combining AND and OR
Write a query to display employees satisfying either of the following conditions:
Department = 'D001' AND Salary > 14000
OR
Department = 'D002' AND Salary > 16000
Use both AND and OR.

15. Using NOT with BETWEEN
Write a query to display employees whose salary is **not between 12000 and 16000**.
Use:
NOT BETWEEN